



# Statistics of Financial Markets

## Seminar 5: Heavy tails and extreme value theory



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## Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ ES: Expected Shortfall
- ▣ EVT: Extreme Value Theory
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GEV: Generalised Extreme Value (distribution)
- ▣ GPD: Generalised Pareto Distribution
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ POT: Peaks Over Threshold
- ▣ QQ: Quantile–Quantile (plot)
- ▣ SE: Standard Error

## Today's Question and Route

- **Question:** how heavy are the tails of daily losses, and how large is the loss we should expect once in 1 000 days or once in 10 years?
  - ▶ this seminar comes **before** Lecture 5: the section “What You Need for Today” gives every definition the tasks use
- Route
  - ▶ Part A: Hill estimates, Pareto tails, GEV return levels and POT risk measures, on paper
  - ▶ Part B: Hill, POT and block maxima on DAX, BET, Bitcoin, BVB stocks and the S&P 500, each with a measure of precision and an interpretation question
  - ▶ Part C: an open question for a project and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section
- Nothing is handed in: the seminar is for practice; the solutions of [Proposed] tasks are discussed in class

## Exercise Map

| Task   | Question                                                                                  | Type                | Model  |
|--------|-------------------------------------------------------------------------------------------|---------------------|--------|
| A1, A2 | the Hill estimator and the Hill quantile from a few large losses                          | Solved,<br>Proposed | A1     |
| A3, A4 | tail probabilities, VaR, ES and mean excess of a Pareto tail                              | Solved,<br>Proposed | A3     |
| A5, A6 | GEV probabilities and return levels                                                       | Solved,<br>Proposed | A5     |
| A7, A8 | VaR and ES from a POT fit, step by step                                                   | Solved,<br>Proposed | A7     |
| B1, B2 | Hill plots and Hill estimates with standard errors for DAX, BET, Bitcoin and BVB stocks   | Solved,<br>Proposed | B1     |
| B3, B4 | POT for the BET and for Bitcoin: EVT vs historical vs Normal; the choice of the threshold | Solved,<br>Proposed | B3     |
| B5, B6 | quarterly maxima of S&P 500 losses; an out-of-sample check of VaR 1%                      | Solved,<br>Proposed | B5     |
| C1, C2 | has the tail of Bitcoin changed? what is wrong in an AI answer?                           | Proposed            | B1, B3 |

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

## Data Used

| Series                                     | Source | Price used           | Period    |
|--------------------------------------------|--------|----------------------|-----------|
| S&P 500, DAX                               | EODHD  | close                | 1990–2026 |
| BET                                        | EODHD  | close                | 1997–2026 |
| Bitcoin                                    | EODHD  | close, 7 days a week | 2014–2026 |
| OMV Petrom (SNP), Banca Transilvania (TLV) | EODHD  | adjusted close       | 2010–2026 |

- Daily log returns in %:  $r_t = 100 (\ln P_t - \ln P_{t-1})$ ; daily losses  $L_t = -r_t$ ; weekends and repeated holiday closes are dropped (except Bitcoin)
- TLV: the days 30 and 31 May 2016 are removed (a price adjustment applied one day late, [Chapter 2](#))
- In the notebook: `losses('dax')`; no account or key is needed

## What You Need for Today (1/5): Heavy Tails

- Tail of the losses:  $\bar{F}(x) = P(L > x)$ ; **heavy (power) tail** with **tail index**  $\alpha$ :  $\bar{F}(x) \approx Cx^{-\alpha}$  for large  $x$ 
  - ▶ doubling the loss divides its probability by  $2^\alpha$ ; on log-log axes: a line with slope  $-\alpha$
  - ▶ light tail (Normal, Exponential):  $\bar{F}(x)$  falls faster than any power
- **Pareto tail**:  $P(L > x) = p_0 (x/x_0)^{-\alpha}$  for  $x \geq x_0$ 
  - ▶ VaR at level  $p < p_0$ :  $\text{VaR}_p = x_0 (p_0/p)^{1/\alpha}$ ;  $\text{ES}_p = \text{VaR}_p \alpha / (\alpha - 1)$  for  $\alpha > 1$
- Moments:  $E|L|^m < \infty$  only for  $m < \alpha$  ( $\alpha \leq 2$ : no variance;  $\alpha \leq 4$ : no kurtosis)

## What You Need for Today (2/5): the Hill Estimator

- Order the losses  $L_{(1)} \geq L_{(2)} \geq \dots$ ; use the  $k$  largest:  $\hat{\alpha}_k = \left[ \frac{1}{k} \sum_{i=1}^k \ln \frac{L_{(i)}}{L_{(k+1)}} \right]^{-1}$  Hill (1975)
  - ▶ i.i.d. (independent, identically distributed) losses:  $SE \approx \hat{\alpha}/\sqrt{k}$ ; 95% CI  $\hat{\alpha}(1 \pm 1.96/\sqrt{k})$
  - ▶ **moving-block bootstrap**: resample blocks of 20 consecutive days; keeps the clusters of large losses
- **Hill plot**:  $\hat{\alpha}_k$  against  $k$ ; choose  $k$  in a flat region; here  $k = 2.5\%$  of  $n$
- **Hill quantile Weissman (1978)**:  $\hat{x}_p = L_{(k+1)} (k/(np))^{1/\hat{\alpha}}$ , the loss exceeded with probability  $p$

## What You Need for Today (3/5): Peaks over Threshold

- **POT** (peaks over threshold): the excesses  $Y = L - u$  over a high threshold  $u$  follow a **GPD** (generalised Pareto distribution) [Pickands \(1975\)](#), [Balkema & de Haan \(1974\)](#)
  - ▶  $P(Y \leq y) = 1 - (1 + \xi y/\beta)^{-1/\xi}$ ; shape  $\xi$  ( $\xi > 0$ :  $\alpha = 1/\xi$ ), scale  $\beta > 0$
  - ▶ **mean excess**  $e(u) = E[L - u \mid L > u]$ ; for a GPD:  $(\beta + \xi(v - u))/(1 - \xi)$ , a rising line
- With  $n$  days and  $N_u$  losses above  $u$ , for  $p < N_u/n$  [McNeil & Frey \(2000\)](#):
  - ▶  $\text{VaR}_p = u + \frac{\beta}{\xi} \left[ \left( \frac{np}{N_u} \right)^{-\xi} - 1 \right], \quad \text{ES}_p = \frac{\text{VaR}_p + \beta - \xi u}{1 - \xi}$
- Threshold:  $u$  = the 90% quantile of the losses (the 10% largest losses, as in [McNeil & Frey \(2000\)](#)); check with the mean excess and the QQ plot of the excesses

## What You Need for Today (4/5): Block Maxima and the GEV

- $M$  = the largest daily loss in a block (a month, a quarter); for long blocks  $M$  follows a **GEV** (generalised extreme value) law [Fisher & Tippett \(1928\)](#), [Gnedenko \(1943\)](#)
  - ▶  $P(M \leq x) = \exp\{-[1 + \xi(x - \mu)/\sigma]^{-1/\xi}\}$ ;  $\xi = 0$ : Gumbel,  $\exp\{-e^{-(x-\mu)/\sigma}\}$
  - ▶  $\xi > 0$  Fréchet (heavy tails),  $\xi = 0$  Gumbel (light),  $\xi < 0$  Weibull (bounded)
- **Return level** for  $m$  blocks: the level exceeded on average once in  $m$  blocks
  - ▶  $z = \mu + \frac{\sigma}{\xi} [(-\ln(1 - 1/m))^{-\xi} - 1]$ ; 10 years = 120 months = 40 quarters
  - ▶ return period of a level  $x$ :  $1/P(M > x)$  blocks

## What You Need for Today (5/5): VaR, ES and Their Checks

- $\text{VaR}_p$ : the loss exceeded with probability  $p$  (VaR 1%, VaR 0.1%);  $\text{ES}_p$ : the mean loss beyond  $\text{VaR}_p$  (ES 2.5%)
  - ▶ historical: the empirical  $(1 - p)$  quantile of the losses and the mean of the losses beyond it
  - ▶ Normal:  $\hat{\mu}_L + \hat{\sigma} z_{1-p}$  and  $\hat{\mu}_L + \hat{\sigma} \varphi(z_{1-p})/p$  ( $\varphi$ : standard Normal density)
- **Out-of-sample check**: estimate VaR on a first period, count the days of a second period with a loss above it
  - ▶ if VaR is right, the count  $X \sim \text{Binomial}(n, p)$ , expected  $np$ ; two-sided p-value  $2 \min\{P(X \leq x), P(X \geq x)\}$
- Quick checks:  $\text{VaR } 0.1\% > \text{VaR } 1\%$ ;  $\text{ES } 2.5\% > \text{VaR } 2.5\%$ ; a GPD with  $\xi < 1/2$  has a finite variance

## A1: Hill from Eight Losses [Solved]

### Task

- ▣ The eight largest daily losses (%) of a stock, in  $n = 2500$  days: 12.4, 9.8, 8.1, 7.5, 6.9, 6.2, 5.8, 5.5.
- ▣ 1. Compute the Hill estimate with  $k = 7$ .
- ▣ 2. Compute its standard error and the 95% interval.
- ▣ 3. Compute the Hill quantile for  $p = 0.1\%$ .
- ▣ Report:  $\hat{\alpha}$ , SE, the interval, VaR 0.1% and one sentence on the moments.

### Solution

- ▣ 1.  $\ln(L_{(i)}/5.5)$ : 0.813, 0.578, 0.387, 0.310, 0.227, 0.120, 0.053; mean 0.355;  $\hat{\alpha} = 2.81$
- ▣ 2.  $SE = 2.81/\sqrt{7} = 1.06$ ; interval  $[0.73, 4.90]$ : very wide with  $k = 7$
- ▣ 3.  $k/(np) = 7/2.5 = 2.800$ ;  $2.800^{1/2.81} = 1.44$ ;  $\hat{x}_{0.001} = 5.5 \times 1.44 = 7.93\%$
- ▣  $\hat{\alpha} \approx 2.8$ : variance finite, kurtosis infinite; but the interval also allows  $\alpha < 2$ .

## A2: Hill for a Crypto Asset [Proposed]

Solution discussed in class

### Task

- ▣ The nine largest daily losses (%) in  $n = 3000$  days: 15.2, 11.0, 9.4, 8.0, 7.7, 7.1, 6.6, 6.3, 6.0. Model: A1.
- ▣ 1. Compute the Hill estimate with  $k = 8$ , its standard error and the 95% interval.
- ▣ 2. Compute the Hill quantile for  $p = 0.1\%$ .
- ▣ 3. Explain why the interval is so wide and what would make it narrower.
- ▣ Report:  $\hat{\alpha}$ , SE, the interval, VaR 0.1% and one sentence.

## A3: A Pareto Tail [Solved]

## Task

- ▣ Daily losses:  $P(L > x) = 2\% \times (x/2.5)^{-3}$  for  $x \geq 2.5\%$ .
- ▣ 1. Compute  $P(L > 5\%)$  and  $P(L > 10\%)$ .
- ▣ 2. Compute VaR 1%, VaR 0.1% and ES 1%.
- ▣ 3. Compute the mean excess at  $u = 4\%$ .
- ▣ 4. Compare  $P(L > 5\%)$  and  $P(L > 10\%)$  with a Normal law  $N(0, \sigma^2)$  that has the same  $P(L > 2.5\%) = 2\%$ .
- ▣ Report: four probabilities, three risk numbers,  $e(4)$  and one sentence.

## Solution

- ▣ 1.  $2\% \times 2^{-3} = 0.250\%$ ;  $2\% \times 4^{-3} = 0.031\%$
- ▣ 2.  $\text{VaR}_p = 2.5 (2\%/p)^{1/3}$ :  $\text{VaR } 1\% = 3.15\%$ ,  $\text{VaR } 0.1\% = 6.79\%$ ;  $\text{ES } 1\% = 3.15 \times 3/2 = 4.72\%$
- ▣ 3.  $e(u) = u/(\alpha - 1) = 4/2 = 2.00\%$
- ▣ 4.  $\sigma = 2.5/z_{0.98} = 1.22$ ; Normal:  $P(L > 5) = 2.0 \times 10^{-3}\%$ ,  $P(L > 10) = 1.1 \times 10^{-14}\%$ : the Normal tail vanishes

## A4: A Heavier Pareto Tail [Proposed]

Solution discussed in class

### Task

- ▣  $P(L > x) = 1.5\% \times (x/3)^{-2.5}$  for  $x \geq 3\%$ . Model: A3.
- ▣ 1. Compute  $P(L > 6\%)$  and  $P(L > 12\%)$ .
- ▣ 2. Compute VaR 0.5%, VaR 0.1%, ES 0.5% and the mean excess at  $u = 5\%$ .
- ▣ 3. Say which moments of  $L$  exist.
- ▣ Report: two probabilities, four numbers and one sentence.

## A5: A GEV Return Level [Solved]

### Task

- ▣ Monthly maxima of daily losses (%) follow a GEV with  $\xi = 0.25$ ,  $\mu = 1.5$ ,  $\sigma = 0.8$ .
- ▣ 1. Compute the probability that the largest daily loss of a month exceeds 6%.
- ▣ 2. Compute the return period of a 6% loss in years.
- ▣ 3. Compute the 10-year return level.
- ▣ 4. Name the type of the GEV and its tail index.
- ▣ Report: one probability, one period, one level and one sentence.

### Solution

- ▣ 1.  $1 + 0.25(6 - 1.5)/0.8 = 2.406$ ;  
 $P = 1 - \exp\{-2.406^{-4}\} = 2.94\%$
- ▣ 2.  $1/P = 34$  months, i.e. about 2.8 years
- ▣ 3.  $m = 120$ :  $-\ln(1 - 1/120) = 0.00837$ ;  $0.00837^{-0.25} = 3.306$ ;  
 $z = 1.5 + (0.8/0.25)(3.306 - 1) = 8.88\%$
- ▣ 4.  $\xi > 0$ : Fréchet, tail index  $1/\xi = 4$ .

## A6: The Same Question with a Gumbel Law [Proposed]

Solution discussed in class

### Task

- ▣ Monthly maxima follow a Gumbel law ( $\xi = 0$ ) with  $\mu = 1.5$ ,  $\sigma = 0.8$ . Model: A5.
- ▣ 1. Compute  $P(M > 6\%)$  and the return period of 6% in years.
- ▣ 2. Compute the 10-year return level,  $z = \mu - \sigma \ln(-\ln(1 - 1/120))$ .
- ▣ 3. Compare with A5: what does  $\xi = 0.25$  change?
- ▣ Report: one probability, one period, one level and one sentence.

## A7: POT Risk Measures, Step by Step [Solved]

### Task

- ▣  $n = 5000$  daily losses;  $u = 2\%$ ;  $N_u = 250$  losses above  $u$ ; GPD fit:  $\xi = 0.2$ ,  $\beta = 0.8$ .
- ▣ 1. Compute VaR 1% and VaR 0.1%.
- ▣ 2. Compute VaR 2.5% and ES 2.5%.
- ▣ 3. Compute the mean excess at  $u$  and the tail index.
- ▣ Report: four risk numbers,  $e(u)$ ,  $\alpha$  and one sentence.

### Solution

- ▣ 1.  $\beta/\xi = 4.00$ ;  $p = 1\%$ :  $np/N_u = 0.200$ ,  $0.200^{-0.2} = 1.380$ ,  $\text{VaR} = 2 + 4(1.380 - 1) = 3.52\%$ ;  $p = 0.1\%$ :  $2.187$ ,  $\text{VaR} = 6.75\%$
- ▣ 2.  $np/N_u = 0.500$ ,  $\text{VaR } 2.5\% = 2.59\%$ ;  $\text{ES} = (2.59 + 0.8 - 0.4)/0.8 = 3.74\%$
- ▣ 3.  $e(u) = \beta/(1 - \xi) = 1.00\%$ ;  $\alpha = 1/\xi = 5$ : a heavy tail, yet with a finite kurtosis.

## A8: POT for a Crypto Asset [Proposed]

Solution discussed in class

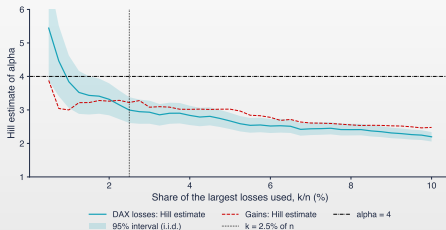
### Task

- ▣  $n = 4000$ ;  $u = 3.1\%$ ;  $N_u = 200$ ; GPD fit:  $\xi = 0.3$ ,  $\beta = 1.5$ . Model: A7.
- ▣ 1. Compute VaR 1%, VaR 0.1% and ES 2.5%.
- ▣ 2. Compute the ratio ES 1% / VaR 1% and compare it with  $1/(1 - \xi)$ .
- ▣ 3. Say whether the formulas can be used for VaR 10%, and why.
- ▣ Report: three risk numbers, one ratio and two sentences.

## B1: How Heavy Is the Tail of DAX Losses? [Solved]

- ▣ **Question:** what is the tail index of daily DAX losses, how precise is it, and does the kurtosis of DAX returns exist?
- ▣ **Inputs:** DAX closes, 1990–2026, daily losses  $L_t = -r_t$  in %
- ▣ **Tasks**
  1. Draw the Hill plot of the losses for  $k$  between 0.5% and 10% of  $n$ , with the i.i.d. 95% band.
  2. Compute  $\hat{\alpha}$  at  $k = 2.5\%$  of  $n$  with the i.i.d. SE, the 95% interval and the moving-block bootstrap SE (blocks of 20 days, 500 samples).
  3. Compute the Hill estimate of the gains with the same  $k$ .
  4. Interpretation: is  $\alpha > 4$  (a finite kurtosis) compatible with the data?
- ▣ **Report:** the chart, four numbers and one sentence
- ▣ **Notebook:** section B1

## B1: Solution [Solved]



- $n = 9\,288$ ,  $k = 232$ ,  $L_{(k+1)} = 2.86\%$ :  $\hat{\alpha} = 3.00$ , SE i.i.d. 0.20, interval  $[2.61, 3.38]$ , SE block 0.20; gains: 3.22
- Between  $k = 1\%$  and  $5\%$  of  $n$  the estimate falls slowly, from 3.84 to 2.67; the gains have a lighter tail
- Interpretation:  $\alpha = 4$  is  $-5.0$  block SE away: the kurtosis of DAX returns does not exist, so the sample kurtosis (6.0) is not a stable number

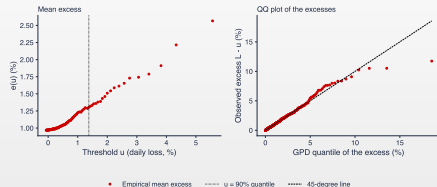
## B2: BET, Bitcoin and Two BVB Stocks [Proposed]

- **Question:** which of the BET, Bitcoin, OMV Petrom and Banca Transilvania has the heaviest tail of losses? Model: B1.
- **Inputs:** BET since 1997, Bitcoin since 2014, SNP and TLV since 2010 (TLV without 30–31 May 2016), daily losses in %
- **Tasks**
  1. Compute  $\hat{\alpha}$  at  $k = 2.5\%$  of  $n$  for each series, with the 95% interval and the block-bootstrap SE.
  2. Add the estimates at  $k = 1\%$  and  $5\%$  of  $n$ .
  3. Test the difference between the BET and Bitcoin with  $z = (\hat{\alpha}_1 - \hat{\alpha}_2) / \sqrt{SE_1^2 + SE_2^2}$ .
  4. Interpretation: is Bitcoin's tail heavier than the BET's once the scale of the losses is set aside?
- **Report:** one table, one  $z$  and two sentences
- **Notebook:** section B2

## B3: POT for the BET [Solved]

- ▣ **Question:** how large are VaR 1%, ES 2.5% and VaR 0.1% of daily BET losses by EVT, compared with historical simulation and the Normal distribution?
- ▣ **Inputs:** BET closes since 1997, daily losses in %;  $u$  = the 90% quantile of the losses
- ▣ **Tasks**
  1. Draw the empirical mean excess function and mark  $u$ .
  2. Fit a GPD to the excesses over  $u$  by maximum likelihood; report  $\hat{\xi}$  with its SE and  $\hat{\beta}$ .
  3. Draw the QQ plot of the excesses against the fitted GPD.
  4. Compute VaR 1%, ES 2.5% and VaR 0.1% by EVT, by historical simulation and by the Normal distribution.
  5. Interpretation: which method would you use for the capital of a bank holding the BET, and why?
- ▣ **Report:** the two charts, a table of nine numbers and two sentences
- ▣ **Notebook:** section B3

## B3: Solution [Solved]



| (%)      | EVT  | historical | Normal |
|----------|------|------------|--------|
| VaR 1%   | 4.44 | 4.33       | 3.39   |
| ES 2.5%  | 4.81 | 4.80       | 3.41   |
| VaR 0.1% | 9.56 | 9.64       | 4.52   |

- $n = 7\,243$ ,  $u = 1.375\%$ ,  $N_u = 725$ ;  $\hat{\xi} = 0.223$  (SE 0.047),  $\hat{\beta} = 1.018$ ; the mean excess rises linearly above  $u$ ; the QQ plot is on the line except the largest excesses
- Interpretation: EVT, which agrees with the data at 1% and gives a smooth VaR 0.1%; the Normal VaR 0.1% is less than half

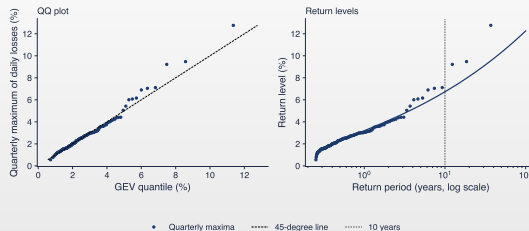
## B4: Does the Threshold Matter? Bitcoin [Proposed]

- **Question:** how much do  $\hat{\xi}$  and the EVT risk measures of Bitcoin change with the threshold? Model: B3.
- **Inputs:** Bitcoin since 2014, daily losses in %;  $u$  at the 85%, 90% and 95% quantiles
- **Tasks**
  1. For each threshold, fit the GPD and report  $u$ ,  $N_u$ ,  $\hat{\xi}$  with its SE and  $\hat{\beta}$ .
  2. Compute VaR 1%, ES 2.5% and VaR 0.1% for each threshold, and by historical simulation.
  3. Interpretation: which quantity is robust to the choice of  $u$ ,  $\hat{\xi}$  or the VaR?
- **Report:** one table and two sentences
- **Notebook:** section B4

## B5: Quarterly Maxima of the S&P 500 [Solved]

- ▣ **Question:** how large is the daily loss that the S&P 500 exceeds once in 10 years, according to a GEV fit to quarterly maxima?
- ▣ **Inputs:** S&P 500 closes since 1990, the largest daily loss of each calendar quarter
- ▣ **Tasks**
  1. Build the quarterly maxima of the daily losses.
  2. Fit a GEV by maximum likelihood; report  $\hat{\xi}$  with its 95% interval,  $\hat{\mu}$  and  $\hat{\sigma}$ .
  3. Draw the QQ plot and the return level plot.
  4. Compute the 10-year return level (40 quarters) and count the quarters above it.
  5. Interpretation: is the S&P 500 tail Fréchet or Gumbel?
- ▣ **Report:** three parameters, one level, the charts and one sentence
- ▣ **Notebook:** section B5

## B5: Solution [Solved]



- 147 quarters:  $\hat{\xi} = 0.21$ , 95% interval  $[0.09, 0.32]$ ;  $\hat{\mu} = 1.97\%$ ,  $\hat{\sigma} = 0.86\%$
- 10-year return level: 6.7%; exceeded in 6 of 147 quarters (3.7 expected in 36.8 years); largest loss 12.8%
- Interpretation:  $\xi = 0$  lies outside the interval: Fréchet, a heavy tail with  $\alpha = 1/\hat{\xi} = 4.8$

## B6: Does VaR 1% Survive a New Period? [Proposed]

- **Question:** estimated until 2014, is the VaR 1% of DAX and BET losses exceeded on about 1% of the days of 2015–2026? Models: B3, B5.
- **Inputs:** DAX since 1990, BET since 1997, daily losses in %; estimation until 31 December 2014, test afterwards
- **Tasks**
  1. Estimate VaR 1% on the first period by EVT (POT,  $u$  at the 90% quantile), historical simulation and the Normal distribution.
  2. Count the test days with a loss above each VaR and compare with  $np$ .
  3. Compute the two-sided binomial p-value of each count.
  4. Interpretation: why can all three methods fail in the same direction?
- **Report:** a table with six counts and p-values, and two sentences
- **Notebook:** section B6

## C1: Has the Tail of Bitcoin Changed? [Proposed]

- ▣ **Question:** is the tail index of Bitcoin losses different in 2014–2017, 2018–2021 and 2022–2026?
- ▣ **Inputs:** Bitcoin daily losses since 2014, split into three periods; models: B1, B2
- ▣ **Tasks**
  1. Compute the Hill  $\hat{\alpha}$  at  $k = 2.5\%$  of  $n$  in each period, with the block-bootstrap SE and a 95% interval.
  2. Say whether any two periods differ significantly.
  3. Propose one change to the method that would give a sharper answer.
  4. Interpretation: what can a sample of about 1 500 days say about a tail index?
- ▣ **Report:** a table, one sentence per comparison and a plan for a project
- ▣ **Notebook:** section C1

## C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to estimate the tail of daily S&P 500 losses since 1990 by POT. The answer:
- (a) With  $u = 1.163\%$ ,  $N_u = 924$  of  $n = 9240$ , the GPD gives  $\xi = 0.158$ , so the tail index is  $\alpha = 0.158$  and the variance is infinite.
- (b)  $\text{VaR } 1\% = u + (\beta/\xi) [(n p / N_u)^\xi - 1] = -0.32\%$ .
- (c)  $\text{ES } 2.5\% = \text{VaR } 2.5\% \times (1 - \xi) = 1.98\%$ .
- (d) The same formula gives a reliable  $\text{VaR } 20\%$ .
- (e) Above  $u$ , the mean excess function of the fitted GPD is a straight line in the threshold.
- Tasks
  - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - ▶ 2. Report: a list of five verdicts with one line of justification each.

## What You Should Take from Today

- ▣ Daily losses have power tails with  $\alpha$  between 2.5 and 3.5: finite variance, no finite kurtosis
- ▣ Hill: fix  $k$  in a flat region of the Hill plot; report a standard error that allows for clustering
- ▣ POT:  $u$  at a high quantile, GPD for the excesses, VaR and ES by closed formulas; check with mean excess and QQ plots
- ▣ Block maxima: GEV, Fréchet for financial losses; return levels beyond the sample are extrapolation
- ▣ An AI answer is a draft: check  $\alpha = 1/\xi$ , the sign of the exponent,  $ES \geq VaR$  and the range  $p < N_u/n$

## After the Seminar

- ▣ Lecture 5 develops each topic of today: crashes, regular variation, Hill, mean excess, GEV, GPD, EVT VaR and ES
- ▣ Try the [Proposed] tasks in the notebook; the solutions are discussed in class
- ▣ C1 can grow into a team project: GARCH-standardised losses, POT with a common threshold, more crypto assets
- ▣ Reading: [Franke, Härdle & Hafner \(2019\)](#), Ch. 18; exercises in [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 18; [McNeil, Frey & Embrechts \(2015\)](#), Ch. 5

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